

Business Memo

To: TechFlow Solutions Management

From: Data Analyst

Date: May 18, 2026

Subject: Staffing recommendation to reduce customer support wait times

TechFlow Solutions is experiencing long customer support wait times that can damage customer satisfaction and brand trust. To better understand the pattern, I generated and analyzed more than 1,000 simulated support call records using Python, Faker, NumPy, pandas, SQLAlchemy, and Streamlit.

The analysis shows that **Monday** has the highest average wait time at **25.9 minutes**. The busiest hours in the dataset are 4:00, 3:00, 13:00, which align closely with normal working hours when customers are most likely to need urgent help. Overall average wait time across all calls is **18.4 minutes**, while Monday and Friday afternoon calls between 1 PM and 5 PM average **37.2 minutes**, confirming the assignment's peak-time hypothesis.

Based on these results, TechFlow should increase staffing by about **40% on Monday and Friday from 1 PM to 5 PM** and slightly expand coverage around the busiest three hours overall. Management should also consider callback options and stronger self-service support for billing and account issues to reduce queue pressure.

If TechFlow applies these changes, the company should be able to reduce support wait times, improve customer experience, and lower the risk of reputational damage from unresolved service complaints. Because this project uses simulated data, the next step should be collecting real call-center data so the staffing plan can be tested and refined.